Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Alford, John Ph.D.		
Project Title:	Formation of Carbon Nanotubes in a Microgravity Environment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jma@tda.com	**Fax:**	FY 303-422-7763
PI Organization Type:	INDUSTRY	**Phone:**	303-940-2304
Organization Name:	TDA Research, Inc.		
PI Address 1:	12345 West 52nd Avenue		
PI Address 2:			
PI Web Page:			
City:	Wheat Ridge	**State:**	CO
Zip Code:	80033	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	03/01/1999	**End Date:**	09/14/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Feikema, Douglas A	**Contact Phone:**	216-433-5707
Contact Email:	Douglas.A.Feikema@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAS3-99142		
Performance Goal No.:	1.22		
Performance Goal Text:	Enable increased combustion system efficiency, reduced pollution, and mitigation of fire risks through insights and data bases obtainable only through microgravity experiments.		
Task Description:	The objective will be accomplished through a series of experiments, which involves nine sub-tasks, and the development of a carbon nanotube reactor that can be dropped at the NASA Glenn 2.2-Second and 5-second drop towers or flown on the KC-135. The information gathered will be used to further understanding of the nanotube formation process and be used to develop better, more economical production processes. The objective of this research is to investigate the effect of buoyancy on the formation mechanisms and morphology (size and length) of single-walled carbon nanotubes. The significance of the work is that by eliminating buoyancy effects from the formation process of SWNTs greater production and manufacturing yields can be obtained. This is currently a major problem and limiting factor in large-scale utilization of the SWNT in aerospace composite materials.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)